

Systematic DA/RT Trading Strategy

Electricity is known to be one of the most volatile commodities on the planet, making its markets difficult to predict. The Electric Reliability Council of Texas (ERCOT) is an independent system operator (ISO) that manages the flow of electricity to over 27 million Texas residents, contributing to 90% of the state's power load. Unlike crude oil and natural gas, electricity cannot be stored efficiently for days, thus supply must match demand at every minute. This dynamic leads to instant rigorous repricing, where small imbalances can result in significant price volatility. The ERCOT market is structured with a Day-Ahead (DA) and Real Time (RT) market. The Day-Ahead market is where power generators and distributors go to submit bids to receive or deliver specific quantities of electricity (usually measured in MW) at a specific time and location the following day. The Real Time market reflects actual grid system conditions, with prices settling every 15 minutes. ERCOT provides traders with the option to submit "virtual bids" which involve financial-only transactions where if your DA bid is cleared, your net P/L is represented by the spread between DA price and RT price. This is commonly known as DA/RT trading. This mechanism is designed to correlate the two markets and provide liquidity by allowing firms that don't possess power assets to operate speculatively. Realizable profit seems trivial by just finding times where $DA < RT$, however, empirical evidence suggests otherwise, shown on page 4. As you can see, from 2022-2025 the DA/RT spread exhibits symmetric dispersion around zero, giving it a mean of approximately 0, but an incredibly high variance. A mean spread of 0 suggests there is no trivial spread arbitrage, and empirically outlines the market's attempt to operate efficiently. However, the high variance creates opportunities for traders to generate alpha. To identify potential sources of alpha in this market, I attempted to collect data on several variables influencing the market and test if combinations of these variables present predictable DART regimes to exploit. In this paper, I will focus on a systematic trading strategy I constructed to exploit conditional inefficiencies in DA/RT spreads using machine learning and risk management.

This research was conducted in a Python Jupyter Lab Notebook, and served to build a model with predictive power on the direction and magnitude of the DA/RT spread. As stated, the observed volatility in electricity markets reveals the relationship between DA/RT levels and its contributing factors are non-linear. Using Python, I constructed a LightGBM model, which is a high performance gradient boosting tree model designed to model these types of non-linear relationships. LightGBM operates by building decision trees on given features, where each subsequent tree is trained to correct the errors of the previous model. The final prediction is formed by aggregating the contributions of all trees, allowing the model to capture complex interactions between the features. To begin constructing this model, I collected the following data from 2022-01-01 -> 2026-01-01: ERCOT RT and DA prices for every 15 minutes, ERCOT hourly load levels, EIA Texas hourly solar output, Houston hourly wind speed, cloud coverage, and temperatures from Open-Meteo's Weather Forecast API, and Henry Hub daily Natural Gas prices from EIA. To make the DA/RT price data compatible with the rest of my DataFrame, I converted them to hourly averages. I uploaded this data into a Pandas DataFrame and then engineered these variables into features for my model to fit on, shown in the table on page 6. To build a systematic DA/RT trading strategy, I constructed a model that predicts the next-hour expected absolute spread represented by $|S_{t+1}| = |RT_{t+1} - DA_{t+1}|$. This model serves as a trade filter to model when the features produce the highest magnitude of P/L. I constructed a second model that attempts to predict the S_{t+1} which determines a short vs. long position. Both models were trained on the data from 2022-2024 and tested on out-of-sample 2025 data to assess predictive performance. The 2025 absolute spread model produced a correlation coefficient of 0.316 and the 2025 directional spread model produced 0.155. The spread magnitude model demonstrates strong predictive power on when large price deviations will occur, while the directional spread model demonstrates moderate predictive power on the spread's direction.

Due to the strength of the magnitude model, in this systematic trading strategy it filters whether or not a trade will be attempted. A trade will only occur if 1) the predicted magnitude of the spread is in the top 10% of all predicted magnitudes and 2) the absolute value of the predicted directional spread is in the top 30% of predicted directional spreads. The second rule serves to eliminate noise, by only filtering for large predicted directional spreads. Together, these rules ensure that trades are only executed in regimes where both the expected direction and magnitude of the DA/RT spread is high. Therefore a position is represented by

$p_t = \text{sign}(s_{t+1}) * 1_{\text{high-opportunity-regime}}$ and the resulting profit and loss is defined by

$PnL_t = p_t(RT_{t+1} - DA_{t+1}) - c * |p_t|$ where c are the estimated transaction costs per trade. In this strategy, a long spread is represented by predicting $RT > DA$ and a short spread is the opposite. All trades in this strategy are virtual orders allowing them to automatically be financially settled at RT_{t+1} .

In a market as volatile as power, risk management plays a very important role in an effective systematic trading strategy. A core concept in risk management is drawdown which is the percentage decline from a trading portfolio's highest peak to its lowest subsequent trough before a new peak is achieved. This metric assesses the worst case scenario of a trading strategy. All trades follow a fixed position size, which is represented by 1MW in my backtests. Under a large capital base (10000MW), this corresponds to allocating proportionally per signal (7%/trade = 70MW), allowing backtest results to be assessed on a per unit basis. I implemented a max drawdown stop that halts trading when max drawdown exceeds a threshold of \$500/MW. I also implemented a rolling loss filter that calculates the total PnL over the last 24 hours, and stops trading if the losses are worse than \$150/MW. These strategies ensure the system is only trading when the signal is strong and the strategy is not in drawdown. I then backtested the systematic trading strategy 4 times on out of sample 2025 DA/RT data, 1) Strategy + Maxdrawdown stop 2) Strategy + Rolling Loss Filter 3) Strategy + both risk management techniques 4) Strategy with no risk management. The results of this backtest are on page 4. All 4 strategies produced very similar equity curves, with consistent gains over the year. The backtest's statistics are on page 4, with the combined equity curve having the best max drawdown. To ensure these results weren't driven by noise, I decided to run another out-of-sample backtest on 2024. I retrained both LightGBM models on 2022-2023 and tested them on 2024. The absolute spread model produced a corr of 0.372 and the directional spread model produced a corr of 0.181, demonstrating my features have strong predictive power when changing the sample and test year. I then backtested all 4 strategies on 2024 DA/RT price data, and got the following results available on page 5. Again, all 4 strategies yielded very similar equity curves with consistent gains over the year. Also, the combined risk management strategy had the lowest max drawdown again, proving to be the best mitigator of risk.

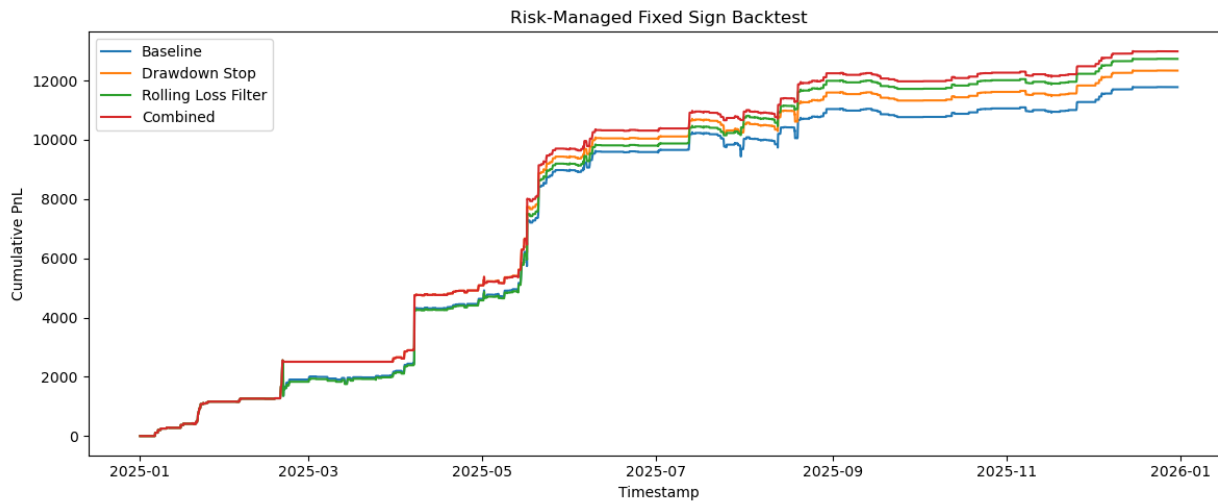
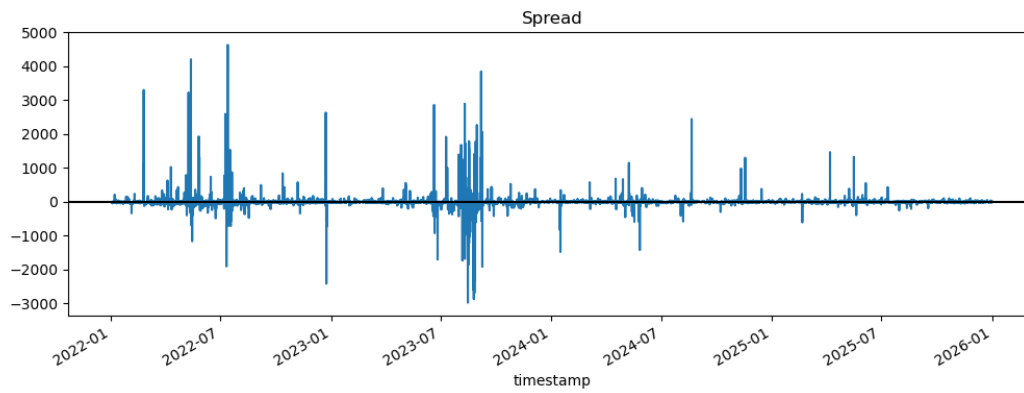
The impact of risk management is implicit in the backtest statistics on pages 4 and 5. In 2025, the baseline strategy had a max drawdown of \$-1200.2125/MW, while the combined risk management strategy had a max drawdown of \$-319.8650/MW, demonstrating significant improvement in downside control. The mean hourly PnL increased from \$1.35/MW to \$1.49/MW, and the hourly Sharpe ratio improved from 0.05 to 0.06. Similarly in 2024, the baseline strategy had a max drawdown of \$-3411.5575/MW, while the combined risk management strategy had a max drawdown of \$-369.5825/MW. Further, the mean hourly PnL increased from \$1.65/MW to \$2.1/MW. These results show not only did risk management improve downside losses, but also increased profitability.

These backtests demonstrated that the two models have strong predictive power on the direction and magnitude of the DA/RT spread. I then needed to test if PnL would hold up under transaction costs. I reran both out-of-sample backtests separately on 2024 and 2025 with a transaction cost of \$1/MW implemented. The results for both are on page 5. In 2025, the strategy produced an annualized Sharpe ratio of 3.24 and 2024 produced an annualized Sharpe ratio of 2.76. The strategy also produced max drawdowns of \$-361.95/MW in 2025 and

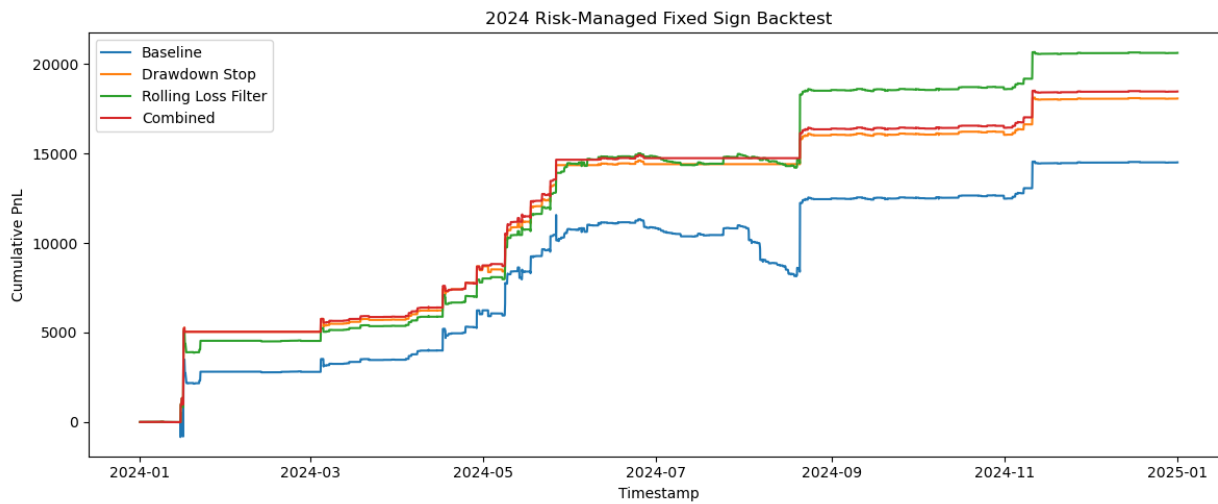
\$-374.58/MW in 2024. This strategy performs consistently across multiple out-of-sample time frames. The results of these backtests indicate the strategy is profitable even after implementing realistic transaction costs. This suggests the models' capabilities are economically meaningful rather than purely statistical. Further, the consistency in results between both out-of-sample backtests firmly supports the models' capabilities and reduces the likelihood that results are due to overfitting.

While the results of this research is compelling, there are limitations to its real world application. The ERCOT market operates like an auction, where traders submit bids and offers across a distribution of prices and quantities in the day-ahead market, and trades only occur if those bids and offers are cleared. The backtests conducted in my research don't simulate this dynamic of the market, and several of the trades made could have not gotten cleared in the day-ahead-market, reducing profitability. Another limitation is the use of hourly average DA/RT prices, when in reality prices clear every at a higher frequency in the day-ahead and real-time markets. As a result, there is a risk my models are not capturing higher frequency stimuli in the market, and are exposed to high intra-hour volatility. However, using higher frequency price data could potentially improve the ability to capture intra hour spread dynamics with more observations, and create more trading opportunities. Another limitation is that both models assume all input variables are known at the time of trade execution, where in some cases, real time data can be discovered late, and variables like weather can contain forecast errors. Another limitation is that this systematic trading strategy assumes sufficient market liquidity to execute trades at desired clearing prices. In reality, large positions can possess significant market impact or be subject to crossing the bid-ask spread, both of which are not modeled in my analysis. Lastly, transaction costs were modeled as a fixed \$1/MW, when in reality costs may vary depending on market conditions.

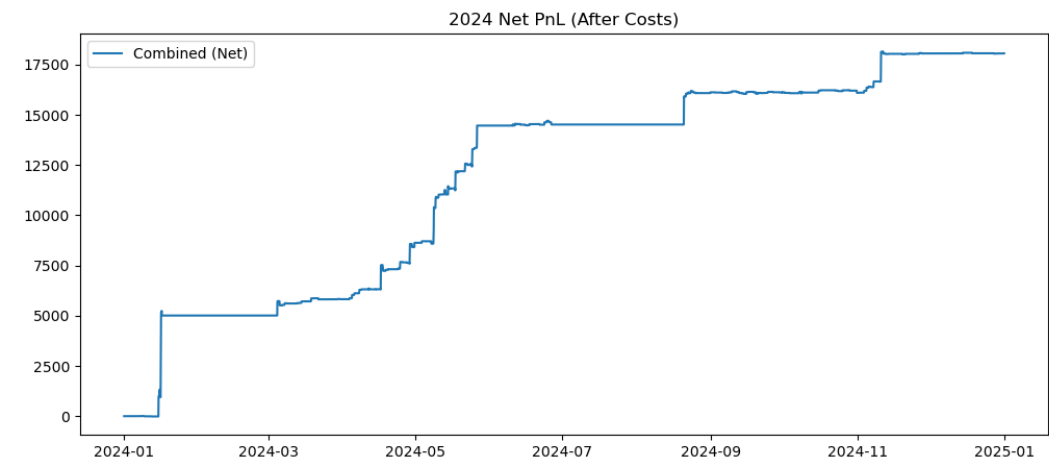
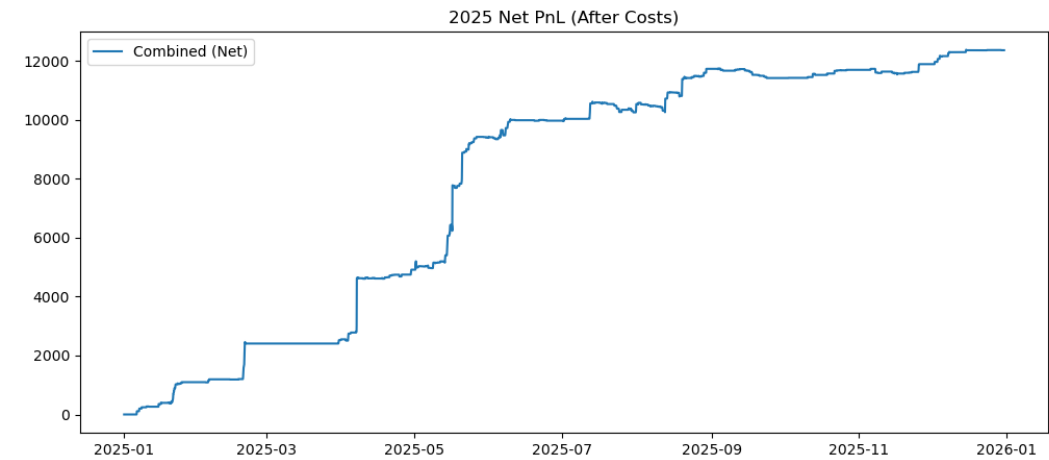
In conclusion, this research demonstrates that while DA/RT spreads in ERCOT nearly possess a mean of 0, significant conditional dynamics exist in high volatility regimes. Through the use of machine learning and simple risk management, this strategy demonstrated the ability to identify and exploit these regimes by generating consistent risk-adjusted returns across multiple out-of-sample periods even after adjusting for estimated transaction costs. While implementing a strategy like this would require more detailed modeling clearing frequencies as well as order clearing dynamics, the findings of this paper provide a strong foundation for building systematic trading strategies in the electricity markets.



Strategy	mean_pnl	std_pnl	hr_sharpe	trades	max_drawdown
Baseline	1.348569	26.877902	0.050174	701.0	-1200.2125
Drawdown Stop	1.412404	24.908187	0.056704	629.0	-483.5175
Rolling Loss Filter	1.457689	26.438572	0.055135	672.0	-1200.2125
Combined	1.486757	24.610115	0.060412	616.0	-319.8650



Strategy	mean_pnl	std_pnl	sharpe	trades	max_drawdown
Baseline	1.651846	53.461398	0.030898	775.0	-3411.5575
Drawdown Stop	2.057873	40.652891	0.050621	421.0	-421.9900
Rolling Loss Filter	2.348170	48.993986	0.047928	691.0	-1256.8475
Combined	2.102616	40.388201	0.052060	405.0	-369.5825



Year	mean_pnl	std_pnl	hourly_sharpe	daily_sharpe	annualized_sharpe	trades	max_drawdown
2024	2.0565043265399066	40.33905024144028	0.05098048452383395	0.17363915027714136	2.7564360569873294	405	-374.58249999999987
2025	1.4162364052661707	24.55522726607279	0.057675556814045105	0.20447518291656258	3.2459428996900046	616	-361.94749999999984

Feature	Feature	Influence on Power Markets
hour	Hour of day (0–23)	Captures intraday demand cycles and price seasonality
dayofweek	Day of week	Weekday vs weekend demand differences
month	Month of year	Seasonal effects (summer cooling, winter heating)
spread_lag1	Prior interval spread	Short-term momentum/mean reversion in spreads
spread_lag24	Spread 24 hrs ago	Daily pattern persistence in spreads
load	Total system demand	Higher load → tighter supply → higher prices
solar	Solar generation output	More solar → lowers net demand → bearish prices
wind	Wind generation output	High wind → suppresses prices due to cheap
gas	Natural gas price	Sets marginal cost for thermal generation
renewables	Total renewable output	Higher renewables → downward pressure on
net_load	Load minus renewables	True demand thermal fleet must meet → key
gas_return_1d	1-day gas price change	Captures fuel cost shocks affecting pricing
gas_dev_7d	Gas deviation from 7-day avg	Identifies abnormal gas pricing regimes
net_load_lag24	Net load 24 hrs ago	Daily demand pattern persistence
spread_vol_24	24-hr spread volatility	Measures regime (calm vs volatile market)
load_lag24	Load 24 hrs ago	Daily load repeatability
temp	Current temperature	Drives heating/cooling demand
temp_lag24	Temp 24 hrs ago	Captures persistence in weather-driven demand
solar_lag24	Solar output 24 hrs ago	Daily solar generation pattern
wind_lag24	Wind output 24 hrs ago	Wind regime persistence
net_load_diff1	Change in net load	Captures ramping stress on grid
net_load_dev_24	Net load vs 24-hr avg	Identifies abnormal demand conditions
spread_mean_24	24-hr avg spread	Mean reversion anchor
hour_sin	Cyclical hour encoding	Helps model smooth intraday seasonality
hour_cos	Cyclical hour encoding	Complements sine for full time cycle

Python Notebook with Constructed DataFrame in .csv format

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